

Learning to fly in formation

A short manual by Natrium.

How to get started

Practise and improve the basics again and again and again, increasing the precision until you master it. If you can't fly solo you can't fly in formation. In this manual I defined 4 phases you can practice, to become better in formation flying.

Important: I wrote this on request in one go. This is just my experience as a simulator pilot, I'm not an actual pilot. Do not use my advice for real world flying.

Phase one – solo flight

Ditch your autopilot

You need to learn flying skills. So even when you fly straight and level, fly without an autopilot. When you're flying straight and level at a defined speed without an autopilot, you're constantly making corrections to keep on track. This means you're constantly training to become better in formation flying. With autopilot on you don't practice.

Practise intentional flying

You cannot fly in a formation when you can't fly intentionally solo.

You need to be able to be in total command of the aircraft. The pitch, yaw, roll, altitude and speed should all be decided by you, and executed that way. That means no climbing or turning or speeding up unless you decide to climb, turn or change speed. If you start a turn at 10,000 feet, you must stay at 10,000 feet halfway during the turn and at the end of the turn. Your roll should be stable.

This means you need to check your instruments and make constant adjustments. The better you are the faster you spot for example your speed dropping and the better you at making the necessary adjustments so you correct it (without overdoing it).

Sloppy pilots allow changes in speed/altitude et cetera, until they become problematic. Good pilots correct any deviation before it becomes a problem. [Watch this Youtube short](#) from a certified flight instructor that drives down this point.

CHALLENGES

Turn on Show path in your Navigation panel, so you can review your flight after flying it.

- 1) Fly straight and level for 5 minutes at 5,000 feet, 250 knots, heading 200
- 2) Speed up to 500 knots without changing your altitude or heading.
- 3) Make a 360 degrees standard turn while staying at the same altitude and speed. Open your Navigation panel, it should show an even circle. At the end of the turn you should be at the same position as when you started the turn.
- 4) Fly a standard pattern: takeoff straight, climb to 2,000 feet. Level off. Turn left 90 degrees. Fly straight for a minute. Turn left another 90 degrees. Descend to 1,000 feet. Turn left

another 90 degrees until you can turn to the heading of the runway. Line up and land.
Review the flight path in the Navigation panel.

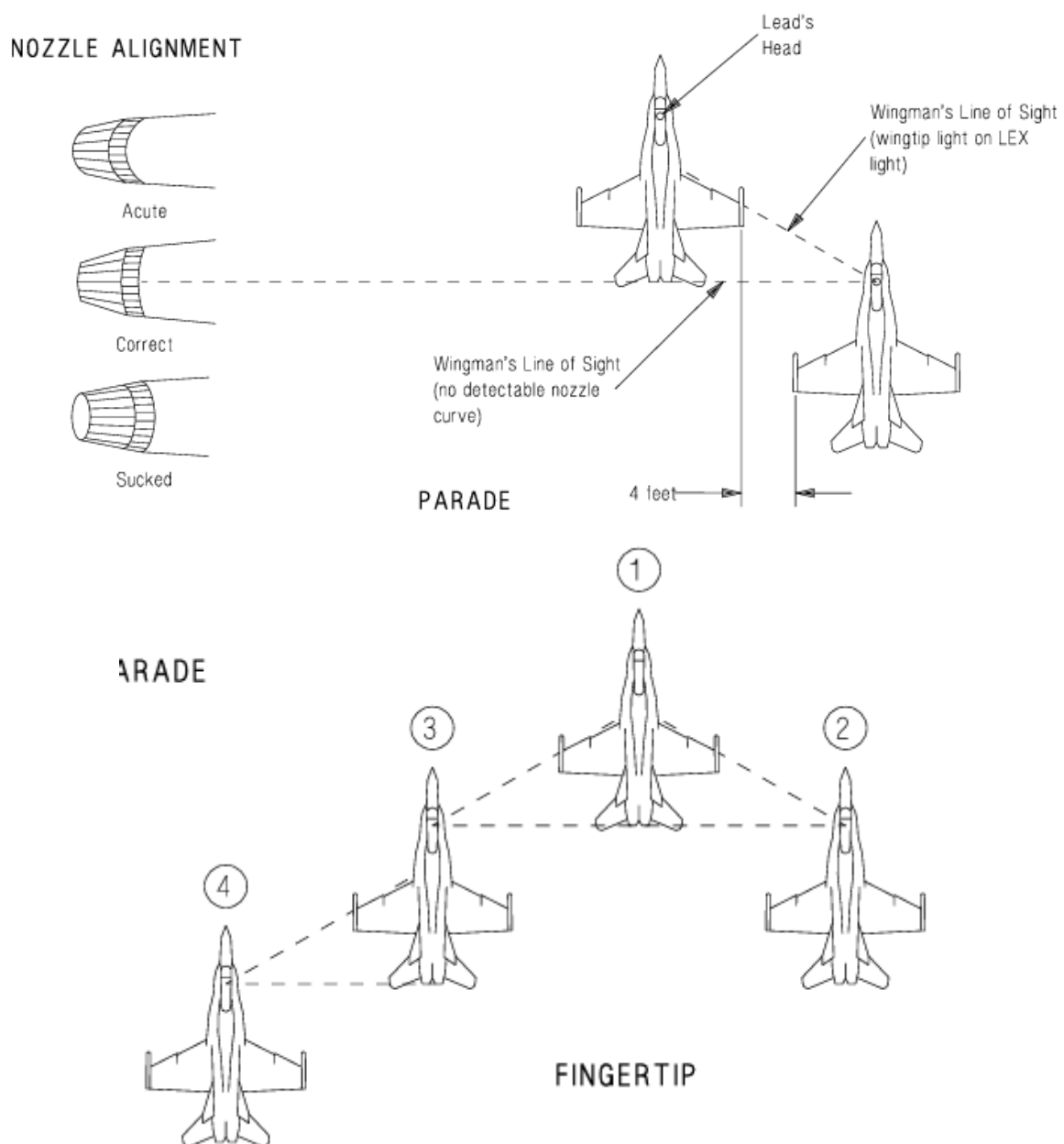
Understanding your aircraft

When you speed up, your wings will generate more lift, so you'll start to nose up. When you roll your aircraft to turn, the wings generate less lift pointing down, so you'll nose down. When you climb, there will be less air, so you'll start to point down. Understand your aircraft so you know how it will respond, and make input changes to correct the tendencies of your aircraft.

For formation flying it's important to understand that your engine takes time to spool up or down. Even though the spool up/down time in GeoFS is fast compared to real jets, there is lag.

Fly in your cockpit

You need to have a fixed point to judge how you fly in relation to the other aircraft. The **only place** where this is possible consistently is in the cockpit.



Fly with a joystick

If you're able, learn to fly with a joystick. They allow you to return your controls to central position easily, allow you to look around while making control inputs with your HAT button, allow you to learn to trim your airplane and allow precise control inputs. A cheap second-hand joystick is way better than using mouse or keyboard, even when you're used to using your mouse.

Trim your aircraft

When you fly with your inputs centered your aircraft should fly in a straight line. You can trim for level flight, or you can trim for ascend or descend. Make a habit to always change your trim whenever your airspeed changes. Increasing your speed means you need to trim down.

Phase two - straight formation flying

Fly in the box

Imagine a box in which your cockpit should stay at all times. The better you become, the smaller the box will be. In the Flight Recorder you can record multiple versions of the same flight as a wingman and watch them all back together. This way you can visualize the size of that box.



About correcting deviations

Your goal is to look like you're fixed in one point in relation to the lead aircraft, as if you're on rails.

You need to make constant and fast corrections. A thing I notice is that people overcorrect when it comes to pitch and roll, and undercorrect with throttle.

Overcorrecting with pitch and roll leads to pilot induced oscillations. Be very smooth with your flight controls. Adjust the sensitivity of your joystick to 0.7. You should be able to let go of your joystick, if that's not the case, you need to trim your aircraft. Having to constantly pull back will create a wobbly flight path and cramped flying. Also retrim your aircraft for a sustained climb or descend.

At the same time with the throttle, I see people using too less throttle adjustments. Especially in multiplayer, the speed of the lead airplane can differ even in straight and level flight, due to lag and how multiplayer works under the hood.

If the speed increase of your target requires an 80% throttle input for the target, anything below 80% will increase your distance to the target. But setting 80% is also not enough, as your reaction time + the time your engine needs to spool up will have increased your distance to the target. So you need to overcompensate for a while (if you're fast, just for a second, do not overdo this) so you start to decrease the distance, and as you start doing that decrease your throttle to stop the overtake and end up at your spot. Just before you reach your spot you should end up at the 80% throttle, so that by the time your engine is stable at 80% you're flying at the same speed at the correct distance. If you overcorrected and closing the distance with your target fast, reduce the throttle for a second to compensate this.

If your target speeds up a slow increase in throttle will only make you fall behind more. So be as fast as possible to react to speed adjustments, and when the speed difference is 10 knots or more, don't mind setting 100% throttle until you are caught up at the same speed.

When you're established on the wing, your corrections will become smaller, but never slower.

Your target is your reference

In the perfect world you fly exactly like your target. When using the same aircraft, if he rolls 5 degrees with 80% power, and you roll 5 degrees with 80% power, you're in sync. So you need to see his settings as a reference point.

If you see your targets settings as your basepoint and make adjustments when you deviate, it doesn't matter whether you're flying straight and level or climbing, speeding up or slowing down. With the same settings you'll both speed up at the same rate.

CHALLENGES

- Fly straight and level as a wingman and reduce the boxsize you're flying in.
- When proficient, move from position (fe. go from right wingman to fly on left).

Phase three – turns and formations

Now that you're able to fly close and steady next to the lead it's time for turns. The bad news is: in GeoFS multiplayer staying close to a turning lead on autopilot is really hard, as the autopilot turns faster than you can flying manual. So when flying close turns should be done flying manual. Also due to lag and a refresh rate of just 0,5Hz, often you're caught by surprise. Luckily in Flight Recorder you don't have this problem.

When you're flying as a wingman, you shouldn't change your reference points on the lead aircraft. This means that when he starts to roll right (and you're flying on his right wing), you need to push your aircraft down.

Do not overdo your turn as a wingman. I often notice myself that I don't want to miss the turn and start too aggressive. Whenever you roll your wing more than the lead aircraft, you'll quickly get off track.

Learn the formations (see below chapter), so you can apply them and know where you need to be if you're flying for example #4.

Phase 4 – Maneuvers

Now that you master the basics, it's time to learn maneuvers. You'll notice that these become way easier now that you're able to fly precise and intentional in close formation.

Use this actual Blue Angels Maneuver Manual to learn the maneuvers:

blueangels.navy.mil/assets/docs/resources/maneuvers-manual.pdf

Further considerations

Flying lead

As a lead you can't fly like you're alone and expect the group to follow you. Here are some rules:

- Maximum 90% throttle, also on take-off (when taking off in formation). Always leave room for your wingman to play with.
- Be VERY gentle on the stick. When turning, first start to roll in slowly before pulling back, so the wingman can push down to get lower and anticipate your turn. When flying in Parade formation (close), every move should be coordinated.

Multiplayer flying

The multiplayer model in GeoFS is limited, to preserve bandwidth. The maximum refreshrate is 0,5 Hz (2 GPS updates per second). That means that if you start a turn there can be half a second a delay before the other notices it. Added to that the autopilot is unrealistic during turns. If you start a turn with the autopilot, the turnrate is almost immediately significant, and more than the rollrate would allow in reality.

Flying in larger groups

- When flying behind another aircraft, fly below their engine wake, never directly behind.
- Flying steady is even more important when flying in bigger groups. If you don't fly steady, but move all over the place, the next airplane that uses you as a reference will have to try to stay close to that mess.

Review your flights

Use the flight recorder to watch different versions of a single flight. Record your first flight and watch it back. Maybe you spot too much vertical movement. Then fly the same flight again and try to improve that aspect. Then watch both recordings back at the same time. Now you can visually identify if you're improving.



Formations

Learn the following formations by name and remember your place in the formation.

We have two ways of flying our formation: in parade (extremely close and demanding), and loose (maintaining more space, just like formations in the military). The parade formation is used during the airshow, the loose formation is used during cruising.

The #1 can issue the following command:

- **“Go parade”** → Move into parade position
- **“Go loose”** → Go to default cruising position.
- **“Close up”** → Close up the formation.
- **“Loosen up”** → Increase the formation distance.

Wingman formation

Fly this when **“Go wingman”** is announced by #1. This is the default formation when flying with two airplanes.



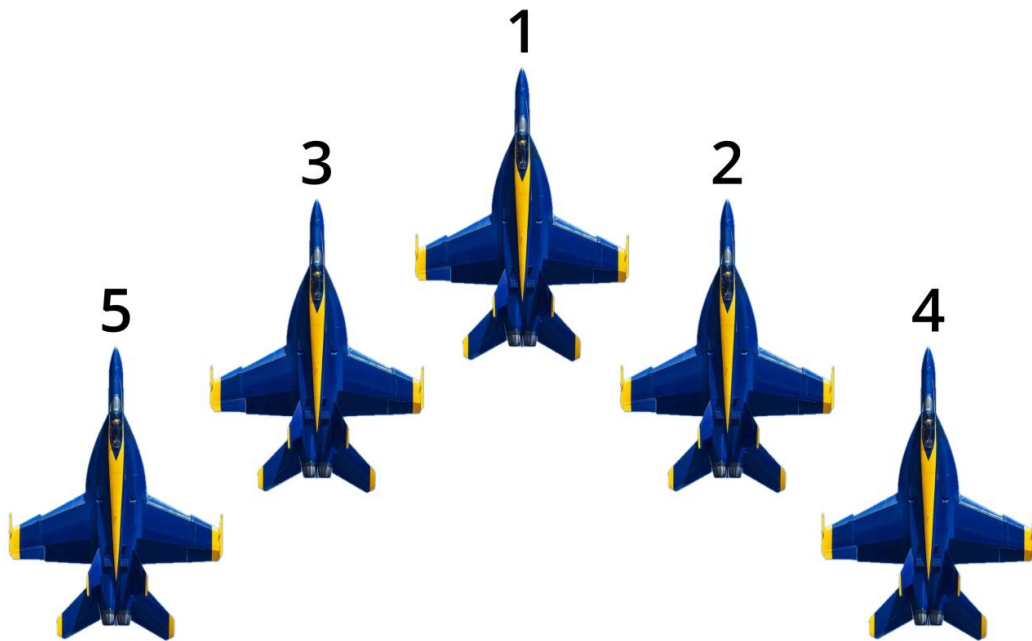
Diamond formation

Fly this when **“Go diamond”** is announced by #1. This is our default 4-ship formation.



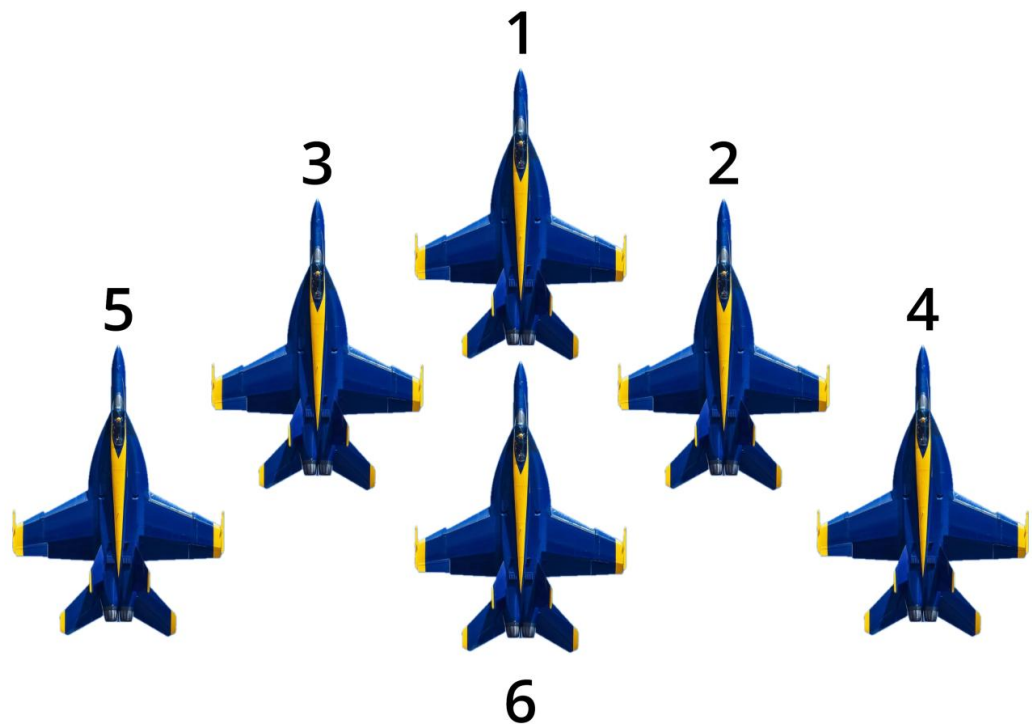
Vic formation

Fly this when **“Go vic”** is announced by #1. This is our default 3-ship and 5-ship formation. Also called a V formation.



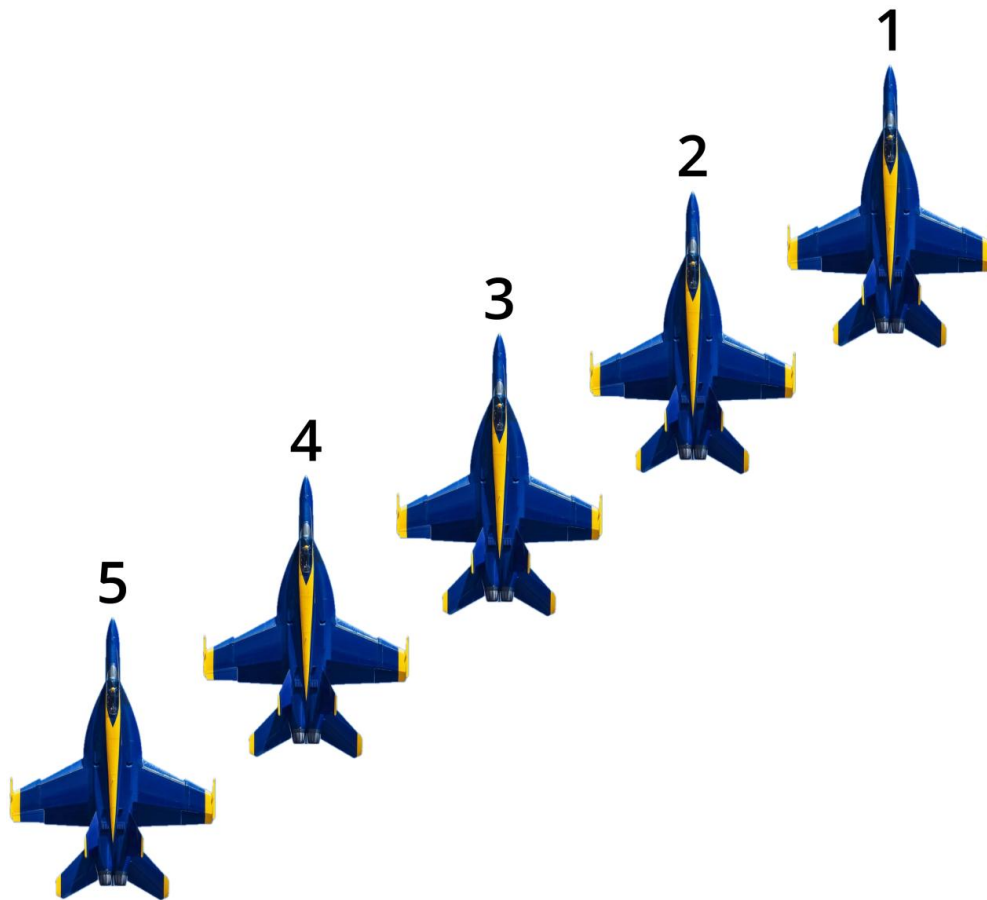
Delta formation

Fly this when **“Go delta”** is announced by #1. This is our default 6-ship formation and is basically the Vic formation, with #6 added in the center.



Echelon formation

Fly this when **“Go echelon”** is announced by #1. By default we fly on the right side of #1, unless **“Go echelon left”** is announced:



Line abreast formation

Fly this when **“Go line”** is announced by #1. By default we fly on the right side of #1, unless **“Go line left”** is announced.



Line astern (trail) formation

Fly this when **“Go trail”** is announced by #1. Fly slightly below the plane in front of you, as in the real world this is needed to prevent turbulence.



Finger four formation

Fly this when **“Go finger four”** is announced by #1. By default we fly a left hand finger four formation, unless **“Go finger four right”** is announced. The finger four is often used in combat, as it can be split easily in two elements (#1 and #2 / #3 and #4).

